Infrastructure Conditions and the Spatial Distribution of Public-Space Offenses in New York City, 2021–2025

Background

Women's experiences in public spaces differ from men's, and the disparity is measurable. For instance, a 2022 UK national survey found that 32% of women felt unsafe in public spaces after dark (Elkin 2022). Similarly, a study of 28 global cities found that women were approximately 10% more likely than men to feel unsafe in metro systems (Ouali et al. 2020). These differences are perceptual; at the same time, they also reflect infrastructural surroundings in public spaces. Infrastructure failures make public space feel unsafe to occupy, thereby restricting women's mobility. Fear of being unsafe undermines women's confidence and limits their social lives (Koskela 1997). This fear is spatially patterned (Valentine 1989). Women frequently manage perceived risk by avoiding "dangerous places" at "dangerous times," so public space becomes temporally segregated. Whether that fear aligns with the actual spatial distribution of offenses and harm, and whether the urban infrastructure conditions in public space are levers on that harm, are empirical questions this analysis takes up. The analysis therefore examines public-space incidents with recorded person victims, and disaggregates them by victim gender, covering 396,876 incidents recorded in NYPD complaint data across 2,243 census tracts (2,228 modeled) from 2021 to 2025.

The analysis has three primary objectives: (1) to characterize the temporal and spatial patterns of public-space offenses over five years; (2) to estimate which infrastructure features and socioeconomic factors are associated with the concentration of offenses across tracts after tweaking spatial autocorrelation, and to examine the correlations between infrastructural features and incidents involving female victims; and (3) to test whether the duration of damaged streetlights is associated with the number of incidents.

Data

"Public space" is operationalized through the recorded premise type, including streets, parks and playgrounds, highways and parkways, bridges, tunnels, bus stops, subway facilities, open lots, and public parking facilities. In this case, offenses are categorized as sexual harm (rape, felony sex crimes, sex crimes), physical harm (felony assault, third-degree assault and related offenses, kidnapping), psychological harm (harassment), and economic harm (robbery, grand larceny, grand larceny of a motor vehicle). Tract-level incidents, therefore, are measured as reported offenses rather than prevalence, and reporting propensity may vary by locations and offense types. An incident is classified as occurring in darkness if its timestamp falls before civil dawn or after civil dusk on its date, computed from solar geometry for New York City. According to this definition, 42.7% of public-space incidents occur in darkness.

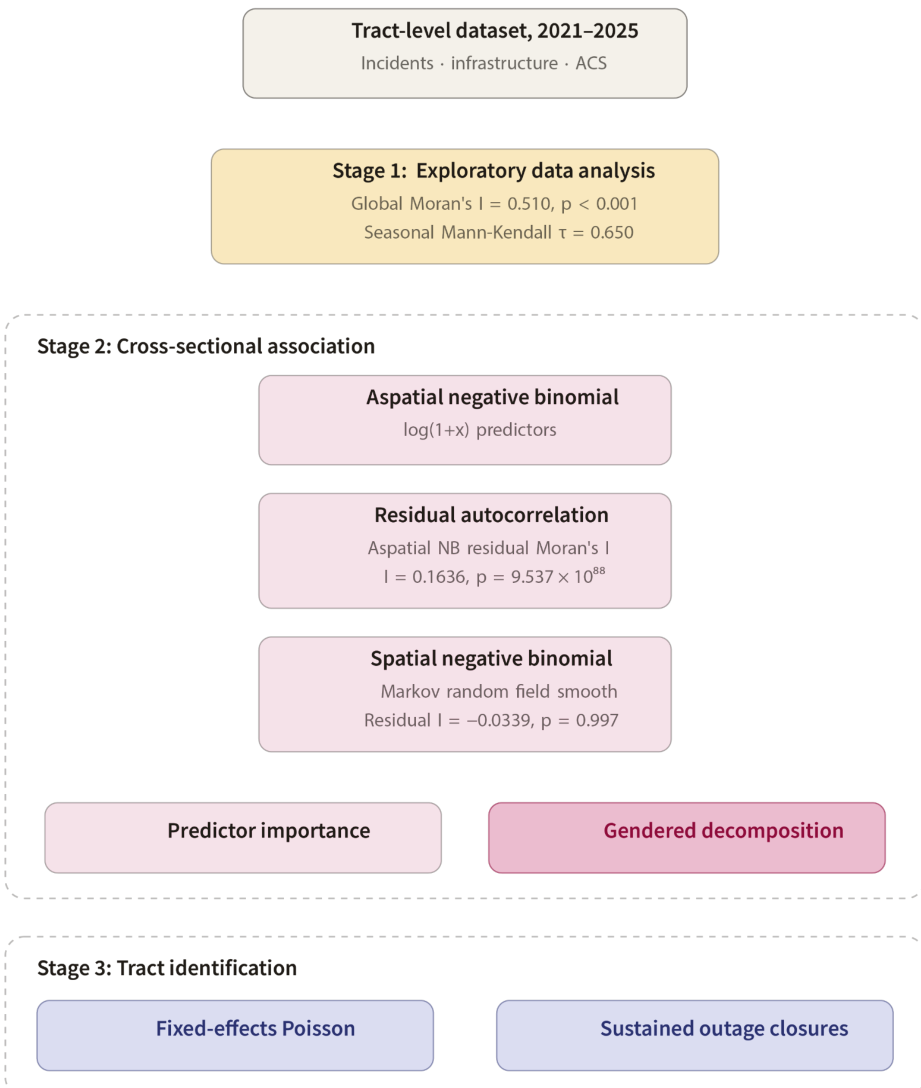
Additionally, infrastructure conditions are measured by six indicators: damaged streetlights (145,598 records from 311), poor pavement (street segments rated 1 to 3 on the Department of Transportation's ten-point scale), broken sidewalks (62,655 reports from 311), transit proximity (distance from each tract centroid to the nearest subway station), vacant land (from PLUTO land-use records), and abandoned or unsecured buildings (6,151 records from Department of Buildings reports via 311).

Socioeconomic indicators from the Census American Community Survey (2020–2024) include the poverty rate, renter share, and housing vacancy rate. All indicators are aggregated at the census tract level. The total resident population serves as the exposure offset in the primary models, while the female population is used for models focusing on female victims.

Method

This analysis proceeds in three stages. First, exploratory spatial data analysis characterizes the distribution of incidents, using global Moran's I, permutation-based LISA, and Getis-Ord Gi* to identify clustering. The seasonal Mann-Kendall test is used to characterize seasonal trends.

Figure 1. Analytical flow chart



Second, cross-sectional negative binomial models are used to estimate associations between infrastructure condition, socioeconomic context, and incident counts, with tract population as the exposure offset. Count-based predictors, which are heavily right-skewed (vacant land skew = 13.6; abandoned buildings skew = 5.0), are entered as standardized log(1 + x) variables. Multicollinearity is low (maximum VIF = 2.99). The aspatial specification leaves substantial spatial structure in its Pearson residuals. Therefore, the preferred model adds a Markov random field smooth over the tract adjacency graph to absorb residual dependence between neighboring tracts. This model is then refit on female-victim and female sexual-offense outcomes to assess whether the correlates of female-victim incidents differ.

Third, a fixed-effects Poisson panel model (2,228 tracts × 59 months; 131,452 tract-months) tests whether within-tract changes in lighting conditions affect offense counts. Each outage is weighted by the number of days it remains open in the month, so a week-long outage carries less weight than a two-month outage, and simultaneous outages are cumulative. The specification includes the current and three lagged months of own-tract darkness and their spatial lags (row-standardized queen adjacency), with tract, year, and calendar-month fixed effects, and standard errors clustered at the tract level. A robustness variant adds monthly 311 flows of sidewalk and building complaints as reporting-propensity controls. An event study around closures of outages lasting at least 30 days then examines incident counts in the months before and after a sustained outage ends.

Spatial and Temporal Patterns

Public-space incidents are unevenly distributed across New York City. They concentrate in Midtown Manhattan and extend into Harlem and the South Bronx (Figure 2), with secondary clusters in Downtown Brooklyn, Jamaica Center and Flushing’s Main Street in Queens, and Coney Island’s commercial district. Strong positive spatial autocorrelation confirms this clustering (global Moran’s I = 0.510, $p < 0.001$), with 368 hotspots and 502 cold spots identified across tracts (Figure 3).

Figure 2: Public-space incidents spatial distribution

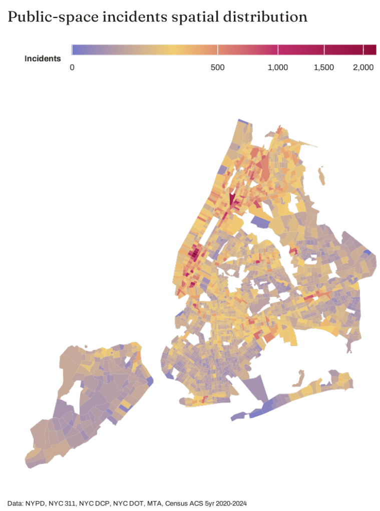
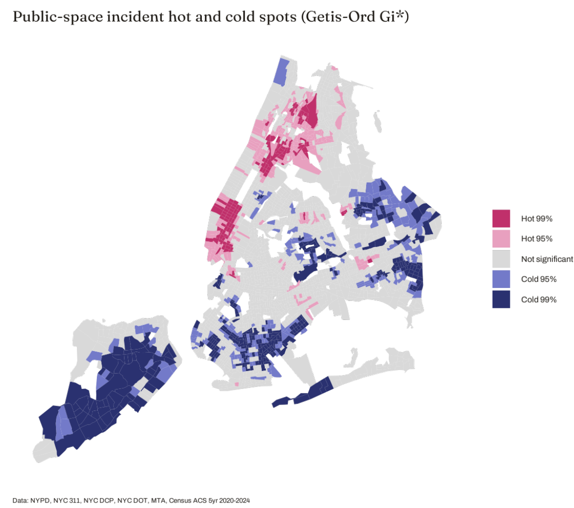


Figure 3: Public-space hot and cold spots



Public-space incident hotspots indicate crime concentration in Manhattan's commercial areas, extending into Harlem and the South Bronx, while cold spots are located in Staten Island, southern Brooklyn, and eastern Queens. Local Moran statistics illustrate how each tract relates to its neighbors, with significant clusters concentrated rather than dispersed (Figures 4 and 5).

Figure 4: Public-space incidents LISA Significance

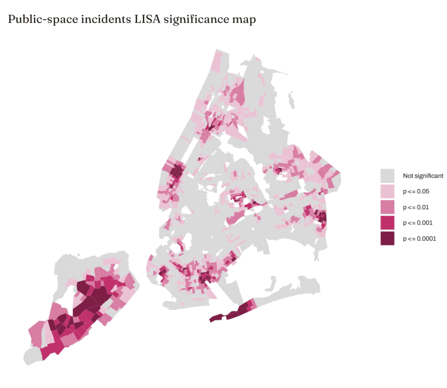
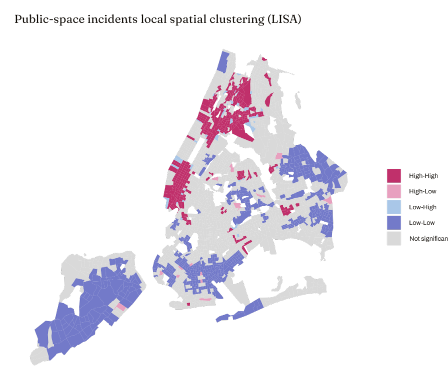
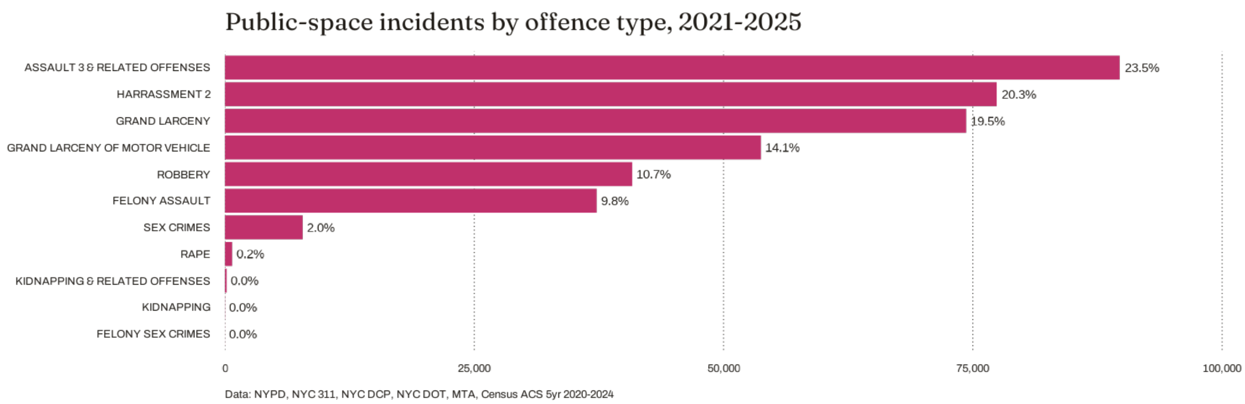


Figure 5: Public-space incidents LISA



The High-High group is a single connected region running down Manhattan's commercial areas from the South Bronx and Harlem to Midtown. The Low-Low group comprises low-density, car-oriented residential areas in Staten Island, eastern Queens, and southern Brooklyn, located away from subways. These neighborhoods also have Pseudo p-values below 0.0001. The few spatial outliers are isolated nodes in quiet surroundings, or in parks and institutional areas. Assault-related offenses constitute the largest category of public-space incidents (Figure 6), accounting for approximately 23.5%. Harassment represents 20.3%, while sex crimes and rape together comprise about 2.2% of all incidents.

Figure 6: Public-space incidents by offense type, 2021-2025



Victim gender varies markedly by offense type (Figure 7). Rape (98% female victims) and sex crimes (82%) predominantly involve female victims, while harassment is also more common among women (59%). The most severe and gendered harms are concentrated in darkness (Figure 8). Rape is the most strongly associated with darkness, occurring in 67% of incidents, followed by robbery (56%) and motor-vehicle theft (51%). Approximately half of felony assault and grand larceny incidents occur in darkness, whereas harassment and kidnapping predominantly occur during daytime.

Figure 7: % female victims by offense type

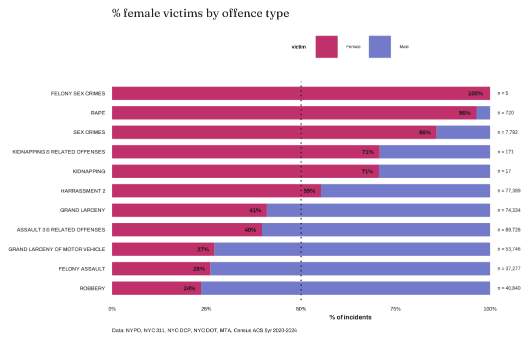
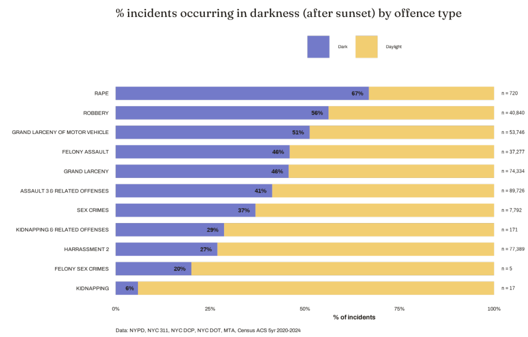


Figure 8: % incidents occurring in darkness



A consistent seasonal pattern is observed over the five-year period. Offenses increase in spring and peak from late May to early October (Figure 9). The upward trend in offenses persists despite this seasonality. The seasonal Mann-Kendall test, which evaluates trends within each calendar month over five years, yields $\tau = 0.650$ ($p < 0.001$), indicating that the entire seasonal profile has shifted upward annually.

Figure 9: Public-space incidents daily trend, 2021-2025

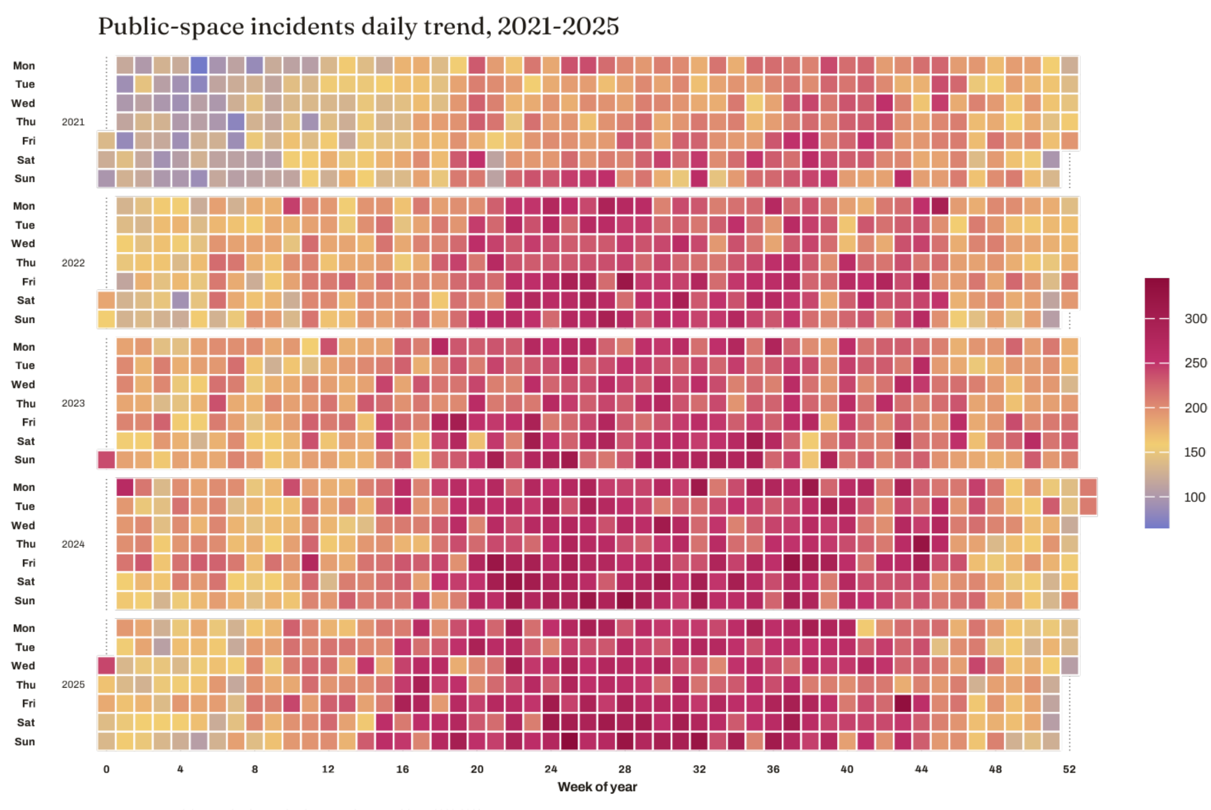
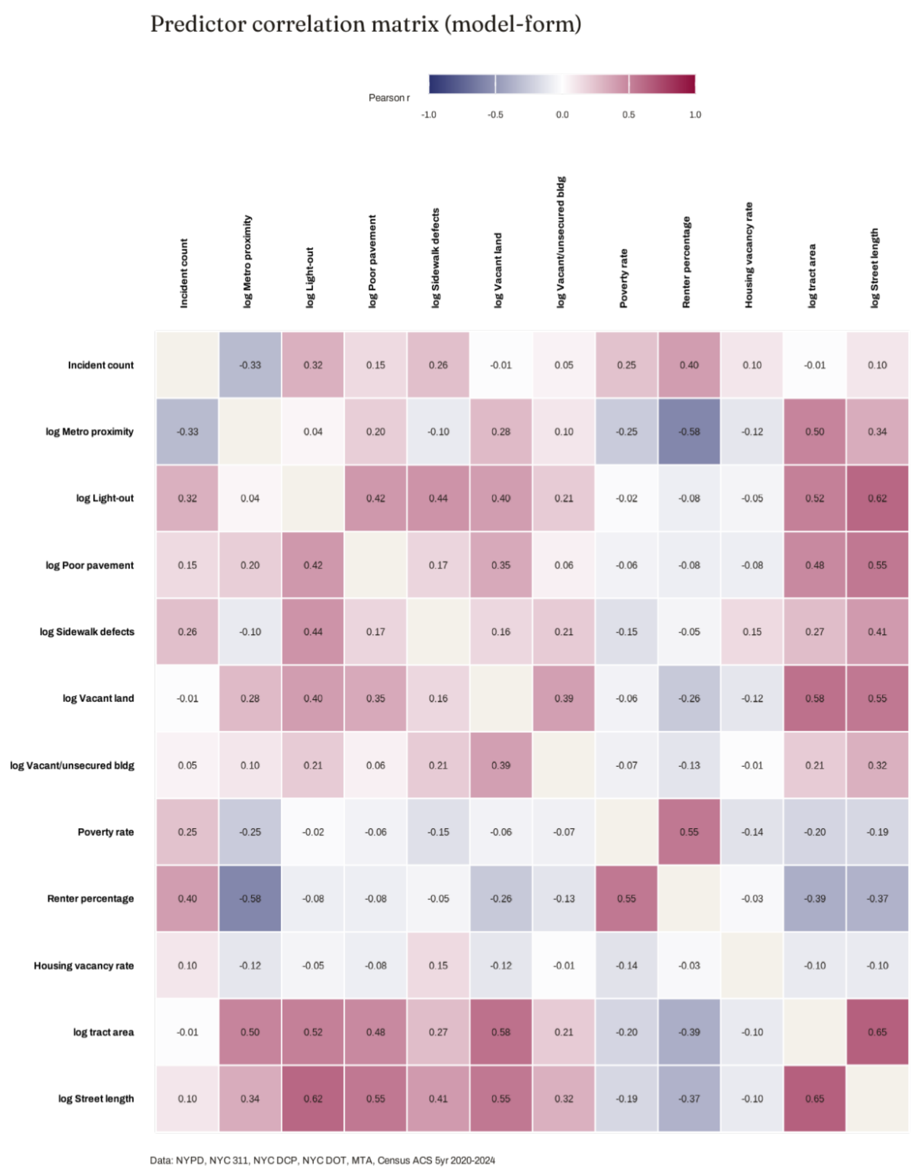


Figure 10 presents pairwise Pearson correlations between the outcome and the log-transformed model predictors. Incident counts correlate most strongly with renter share (r = 0.40), streetlight outages (r = 0.32), and subway distance (r = −0.33). Correlations with deterioration indicators are weaker (e.g., sidewalk defects, r = 0.26) or negligible (e.g., vacant land, r = −0.01). Two aspects of the predictor set indicate that these bivariate associations are confounded.

Figure 10: Predictor correlation matrix (model-form)



First, infrastructure complaints increase with a tract's physical size. Streetlight outages correlate with street length ($r = 0.62$) and tract area ($r = 0.52$), while poor pavement ($r = 0.55$) and vacant land ($r = 0.55$) show similar patterns. Thus, a tract may record many outages due to its extensive street network rather than poor maintenance. Second, renter share is strongly correlated with subway distance ($r = -0.58$) and poverty rate ($r = 0.55$), indicating that its association with incidence partly reflects transit accessibility rather than tenure composition.

Results

Tract-level incident counts are severely overdispersed (mean = 166.7, variance = 27,621.8; variance-to-mean ratio = 166). A likelihood-ratio test rejected the Poisson model in favor of the negative binomial ($\chi^2 = 177{,}706$, $p < 0.001$; AIC 205,044 versus 27,340). A quadratic poverty specification was preferred over a linear one (AIC 27,340 versus 27,431). The aspatial negative binomial model failed the residual diagnostic, with Pearson residuals that were spatially autocorrelated (Moran's $I = 0.164$, $p < 0.001$).

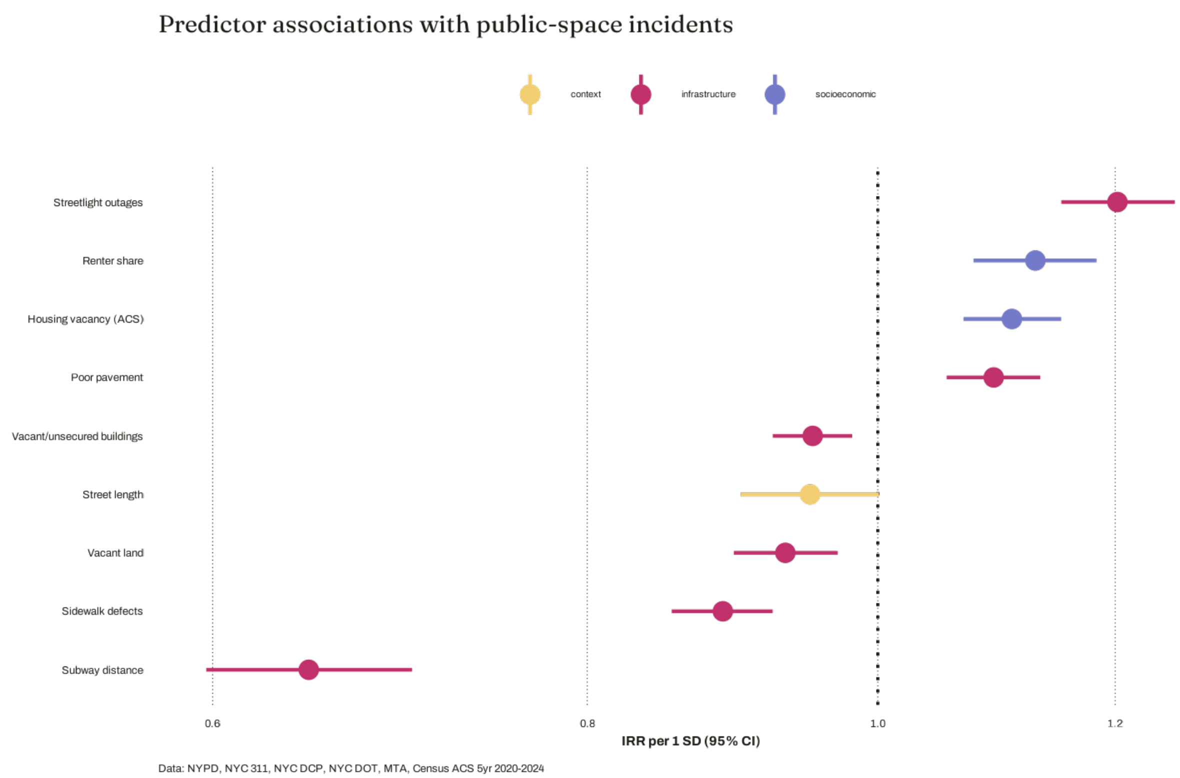
Applying a Markov random field smooth over the tract adjacency graph eliminated spatial dependence. The spatial model's conditional residuals showed no remaining autocorrelation ($I = -0.034$, $p = 0.997$), deviance explained increased to 73.9%, and the AIC improved from 27,340 to 25,643. All subsequent estimates are derived from the spatial model and are expressed as incidence rate ratios (IRRs) per one standard deviation of each log-transformed, standardized predictor.

Figure 11 presents the estimated rate ratios with 95% confidence intervals. Subway distance showed the largest association in magnitude (IRR 0.65 per standard deviation, 95% CI 0.60–0.70, corresponding to a 1.54-fold difference in incident rate per standard deviation), while streetlight outage burden showed the largest positive association (IRR 1.20, 1.15–1.26).

Renter share (IRR 1.13), housing vacancy (1.11), and poor pavement (1.09) were also positively associated with incident rates (all $p < 0.001$).

Indicators of deterioration were either negatively associated with incident rates or showed no association. Sidewalk defects had an IRR of 0.89 ($p < 0.001$); vacant land, 0.93 ($p < 0.001$); vacant and unsecured buildings, 0.95 ($p = 0.001$); and street length, 0.95 ($p = 0.054$). Poverty showed no linear gradient ($p = 0.17$) but showed a weak convex term ($p = 0.011$).

Figure 11: Predictor associations with public-space incidents



The female-victim model, estimated using the resident female population as exposure, produced coefficients similar in sign and magnitude to those of the all-victim model (Table 1), with the largest difference across the eight predictors being 0.027 (sidewalk defects). As the female-victim outcome is nested within the total, the comparison is descriptive, and no test of coefficient equality was conducted. The female sexual-offense model returned no coefficient distinguishable from zero. Sexual offenses accounted for approximately 2% of incidents, and the confidence intervals were correspondingly wide. The streetlight's point estimate (0.345) was the largest among the three models, but its interval included zero.

Table 1. Spatial NB coefficients (log IRR per 1 SD) across the three outcomes

Predictor	**All victims**	**Female victims**	**Female sexual offenses**
Subway distance	−0.437***	−0.442***	−0.197
Streetlight outages	0.184***	0.197***	0.345
Poor pavement	0.089***	0.087***	0.373
Sidewalk defects	−0.119***	−0.092***	0.082
Vacant land	−0.071***	−0.076***	−0.612
Vacant/unsecured buildings	−0.050**	−0.049**	0.214
Street length	−0.052	−0.077*	0.720
Renter share	0.121***	0.124***	0.251
Housing vacancy	0.103***	0.126***	0.095

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Female-victim models use the resident female population as exposure. Tract area and the quadratic poverty terms are included in all models but omitted from the table. Columns are descriptive comparisons because the outcomes are nested.*

The pooled distributed-lag model, estimated without tract fixed effects, returned positive and significant coefficients for all four own-tract darkness lags (contemporaneous $\beta = 0.050$, $p < 0.001$). With tract, year, and calendar-month fixed effects, no own-tract coefficient differed from zero (contemporaneous $\beta = 0.001$, $p = 0.74$; Figure 12).

Within-tract variation in darkness (mean within-tract SD = 1.34) was similar to between-tract variation (SD of tract means = 1.26). Incorporating monthly 311 complaint flows as reporting-propensity controls left all estimates essentially unchanged. The contemporaneous spatial lag remained significant across all specifications ($\beta = 0.024$, $p < 0.001$; $\beta = 0.023$ with controls), with a smaller coefficient at the third lag ($\beta = 0.018$, $p = 0.007$) and null coefficients at the first and second lags.

In the event study, closures of outages lasting at least 30 days (8.0% of tract-months) showed no association with incident counts at either the lead or any of the four event-time terms (all $p > 0.14$; Figure 13). Given cluster-robust standard errors of approximately 0.0065, the study design could exclude closure effects larger than approximately ±1.3% per month at the 95% confidence level.

Figure 12: Effect of dark lamp-days on incidents by lag(FE Poisson)

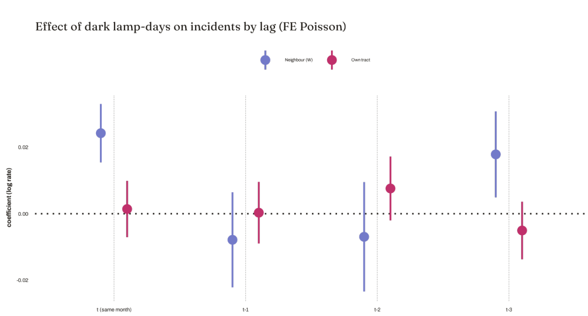
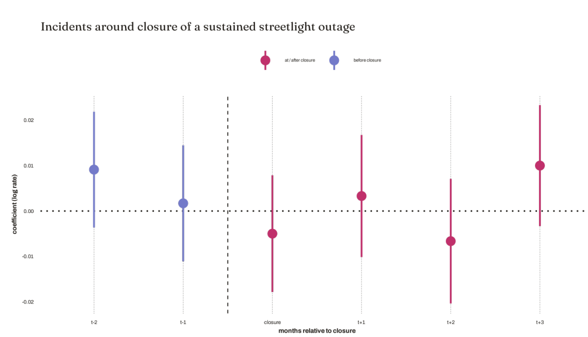


Figure 13: Incidents around closure of a sustained streetlight outage



Conclusion

This analysis described how public-space incidents are distributed in New York City and found that infrastructure and socioeconomic factors are correlated after spatial correction. The 396,876 recorded incidents are strongly clustered (Moran's I = 0.510), have increased over five years, and the most severe and female-victim offenses are concentrated in darkness. Incidence is shaped by activity and transit proximity. Among physical conditions, only streetlight outages and poor pavement are positively associated with incident rates, while other indicators show no significant association. Temporary darkness has no detectable effect within tracts, and monthly effects larger than ±1.3% are excluded. The association between streetlighting and incidents is attributable to long-term differences between tracts.

The environmental factors linked to female victims are the same as those for overall victims, so avoiding certain places only addresses part of the risk women face. What stands out as gendered in these data is the timing and type of offenses. Crimes that mostly affect women also happen most often in darkness. Then the risk after dark and the importance of infrastructure that improves visibility at night, fall mainly on women. Fixing infrastructural problems should not be expected to reduce incident counts the next month. To test these conclusions further, future research will need data on the number of people present, more detailed spatial units for offenses, and studies of long-term infrastructure changes that fall outside this panel's scope.

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